

Alan Lin

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EXPERIENCE

Red Hat

May 2026 - Aug. 2026

Software Engineer Intern - Ecosystems AI

Boston, MA

- Built **Playwright** E2E test suite validating multi-turn **AI agent** conversations through Zammad ticketing UI; asserting metadata transitions across **4+** turns with retry logic and race condition handling
- Developed **Helm chart** for OpenShift AI's Gen AI Studio registering **6 MCP servers** and vector stores; automating RAG ingestion of 10 runbooks via post-install **Kubernetes Job** with readiness probes
- Improved telecom anomaly detection F1 by **48%** and root cause analysis accuracy from 41% to **92.7%** by training **Mantis** and **TimesNet** encoders in **PyTorch**; tuning class weights and focal loss tracked in **MLflow**

Subletr

May 2025 - Aug. 2025

Fullstack Engineer Intern

Boston, MA

- Developed secure authentication system with **JWT** tokens and role-based access control using **Next.js** and **Prisma**; serving **600+** users across **15+** universities and achieving **99.9% uptime**
- Built real-time messaging system with Pusher **Websocket** technology and **MongoDB** integration; reducing message latency by **60%** to enable instant renter-subletter communication
- Designed scalable backend architecture with **REST APIs** and database optimization using **AWS** services; supporting **100+** daily queries with **40% faster** load times

BU Spark!

Jan. 2025 - May 2026

Software Engineer Intern

Boston, MA

- Modernized **open source** biological data platform with **PHP** and **MySQL** batch workflows; enabling **50,000+** genetic sample analysis for research institutions
- Enhanced open source data pipeline by refactoring batch algorithms and image handling; accelerating genomic uploads by **35%** for time-sensitive research
- Migrated **4+** critical features using agile practices and **CI/CD** workflows; maintaining platform stability for researchers while aligning with upstream repository standards

PROJECTS

Boston Police Index | *NextJS, PostgreSQL, GraphQL*

May 2025 - Apr. 2026

- Architected civic transparency platform processing **10,000+** police records with **React**, **GraphQL**, and **PostgreSQL**; empowering journalists and advocates with real-time search and export for accountability research
- Developed interactive visualizations with **D3.js**, **GraphQL APIs**, and **Material-UI**; enabling compensation analysis across **15+** departments for public oversight

Open Source System for 3D Printing Robot | *NextJS, FastAPI, MySQL*

Jan. 2025 - May 2025

- Engineered full-stack web application using **Next.js** and **FastAPI**; accelerating material science experiments through real-time 3D model generation for **10+** researchers across universities
- Created authentication system with Clerk and **MySQL/Prisma**; building **RESTful** APIs that reduced experiment tracking time by **75%** for research teams

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, Java, SQL, Kotlin, C/C++, Bash, HTML/CSS

Frameworks & Libraries: React.js, Next.js, Node.js, FastAPI, Flask, GraphQL, Material-UI, Scikit-Learn, Pandas

Cloud & DevOps: Docker, Kubernetes, OpenShift, AWS, Azure, Linux, CI/CD (GitHub Actions)

Databases & Tools: PostgreSQL, MySQL, MongoDB, Prisma ORM, Firebase, Git, Postman, Jest

EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

- **Relevant Coursework:** Cloud Computing, Distributed Systems, Database Systems, Software Engineering, Web App Development, Machine Learning, Algorithms, HCI & HAI
- **Activities:** BU Spark!, Hack4Impact, BostonHacks, Chinese Student Association